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DDL, DML, DCL, TCL COMMANDS

1. CREATING A TABLE bank_transaction

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator:.....

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information:.....

No object selected

Query 1 SQL File 4* x SQL File 4*

```
269 CREATE TABLE bank_transactions (  
270     txn_id INT PRIMARY KEY,  
271     customer_name VARCHAR(50),  
272     branch_name VARCHAR(50),  
273     transaction_type VARCHAR(20),  
274     amount DECIMAL(10,2),  
275     transaction_date DATE  
276 );  
277 DESCRIBE bank_transactions;
```

Result Grid

	Field	Type	Null	Key	Default	Extra
▶	txn_id	int	NO	PRI	NULL	
	customer_name	varchar(50)	YES		NULL	
	branch_name	varchar(50)	YES		NULL	
	transaction_type	varchar(20)	YES		NULL	
	amount	decimal(10,2)	YES		NULL	
	transaction_date	date	YES		NULL	

Result 1 x Read Only

Output

Action Output

#	Time	Action	Message
✗	1 17:58:27	CREATE TABLE bank_transactions (txn_id INT ...	Error Code: 1046. No database se
✓	2 17:58:34	USE bookflow_db	0 row(s) affected
✓	3 17:58:40	CREATE TABLE bank_transactions (txn_id INT ...	0 row(s) affected
✓	4 17:59:07	DESCRIBE bank_transactions	6 row(s) returned

2.USING ALTER TABLE AND COLOUMN

I added acc.no coloumn

MySQL Workbench

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' tab is selected, showing 'Schemas' and 'Information' sections. The 'Information' section shows 'No object selected'.

The main window shows a SQL script in the 'Query 1' tab. The script contains the following queries:

```
274     amount DECIMAL(10,2),
275     transaction_date DATE
276 );
277 • DESCRIBE bank_transactions;
278
279 • ALTER TABLE bank_transactions
280   ADD account_no VARCHAR(20);
281
282 • DESCRIBE bank_transactions;
```

The 'Result Grid' shows the structure of the 'bank_transactions' table:

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

The 'Output' section shows the 'Action Output' table:

#	Time	Action	Message
1	17:58:27	CREATE TABLE bank_transactions (txn_id INT ...	Error Code: 1046. No database selected
2	17:58:34	USE bookflow_db	0 row(s) affected
3	17:58:40	CREATE TABLE bank_transactions (txn_id INT ...	0 row(s) affected
4	17:59:07	DESCRIBE bank_transactions	6 row(s) returned
5	18:04:01	ALTER TABLE bank_transactions ADD account_n...	0 row(s) affected Records: 0 Duplicate
6	18:06:15	DESCRIBE bank_transactions	7 row(s) returned

3.USING ALTER MODIFY

Modified customer name VARCHAR to 100 which was 50 previously

MySQL Workbench

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Local instance MySQL80 x
- Menu:** File, Edit, View, Query, Database, Server, Tools, Scripting, Help
- Query Editor:** Contains SQL queries for Query 1:

```
277 • DESCRIBE bank_transactions;
278
279 • ALTER TABLE bank_transactions
280 ADD account_no VARCHAR(20);
281
282 • DESCRIBE bank_transactions;
283 • ALTER TABLE bank_transactions
284 MODIFY customer_name VARCHAR(100);
285 • DESCRIBE bank_transactions;
```
- Result Grid:** Displays the structure of the bank_transactions table.

	Field	Type	Null	Key	Default	Extra
▶	txn_id	int	NO	PRI	NULL	
	customer_name	varchar(100)	YES		NULL	
	branch_name	varchar(50)	YES		NULL	
	transaction_type	varchar(20)	YES		NULL	
	amount	decimal(10,2)	YES		NULL	
	transaction_date	date	YES		NULL	
	account_no	varchar(20)	YES		NULL	
- Output:** Shows the execution results of the queries.

#	Time	Action	Message
✓ 3	17:58:40	CREATE TABLE bank_transactions (txn_id IN...	0 row(s) affected
✓ 4	17:59:07	DESCRIBE bank_transactions	6 row(s) returned
✓ 5	18:04:01	ALTER TABLE bank_transactions ADD account_...	0 row(s) affected Records: 0 Duplicates: 0 Wa
✓ 6	18:06:15	DESCRIBE bank_transactions	7 row(s) returned
✓ 7	18:07:55	ALTER TABLE bank_transactions MODIFY custo...	0 row(s) affected Records: 0 Duplicates: 0 Wa
✓ 8	18:08:12	DESCRIBE bank_transactions	7 row(s) returned

4.USING ALTER TABLE RENAME COMMAND

Name of the table is changed

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 4* x SQL File 4*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Result Grid

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

Result 4 x Read Only

Output

Action Output

#	Time	Action	Message
7	18:07:55	ALTER TABLE bank_transactions MODIFY custo...	0 row(s) affected Records: 0 Dupl
8	18:08:12	DESCRIBE bank_transactions	7 row(s) returned
9	18:10:52	RENAME bank_transactions TO customer_transa...	Error Code: 1064. You have an em
10	18:13:16	ALTER TABLE bank_transactions RENAME TO ...	0 row(s) affected
11	18:13:36	DESCRIBE bank_transactions	Error Code: 1146. Table 'bookflow,
12	18:13:56	DESCRIBE customer_transactions	7 row(s) returned

Object Info Session

5.INSERTED VALUE IN TABLE

MySQL Workbench

The screenshot shows the MySQL Workbench interface with a local instance of MySQL 8.0. The left sidebar contains navigation panels for Management, Instance, and Performance. The main window displays a SQL query in the 'Query 1' tab. The query includes an ALTER TABLE statement to rename 'bank_transactions' to 'customer_transactions', followed by DESCRIBE statements and an INSERT INTO statement. The 'Result Grid' shows the execution results of the query, including the schema of 'customer_transactions' and the output of the INSERT and SELECT statements. The 'Output' panel at the bottom shows the execution log with error messages for the first two INSERT attempts and success messages for the third INSERT and the subsequent DESCRIBE and SELECT statements.

Query 1

```
286
287 • ALTER TABLE bank_transactions
288   RENAME TO customer_transactions;
289 • DESCRIBE bank_transactions;
290 • DESCRIBE customer_transactions;
291
292 • INSERT INTO customer_transactions(txn_id,customer_name,branch_n
293   ('1','MADHURIMA','KPHB','CASH','500','2026-08-10');
294 • SELECT *FROM customer_transactions;
```

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶ 1	MADHURIMA	KPHB	CASH	500.00	2026-08-10	
*	NULL	NULL	NULL	NULL	NULL	NULL

Output

#	Time	Action	Message
✗ 16	18:19:42	INSERT INTO customer_transactions(customer_n...	Error Code: 1292. Incorrect date value:
✗ 17	18:20:38	INSERT INTO customer_transactions(customer_n...	Error Code: 1364. Field 'txn_id' doesn't h
✓ 18	18:21:40	INSERT INTO customer_transactions(txn_id,cust...	1 row(s) affected
✗ 19	18:21:58	DESCRIBE customer_transaction	Error Code: 1146. Table 'bookflow_db.c
✓ 20	18:22:11	DESCRIBE customer_transactions	7 row(s) returned
✓ 21	18:22:40	SELECT *FROM customer_transactions LIMIT 0, ...	1 row(s) returned

That value is deleted when I used truncate

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1 SQL File 4* SQL File 4*

```
292 • INSERT INTO customer_transactions(txn_id,customer_name,branch_n
293 ('1','MADHURIMA','KPHB','CASH','500','2026-08-10');
294 • SELECT *FROM customer_transactions;
295 • TRUNCATE TABLE customer_transactions;
296 • DESCRIBE customer_transactions;
297
298
299
300
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

Result 9

Read Only Context Help Snippets

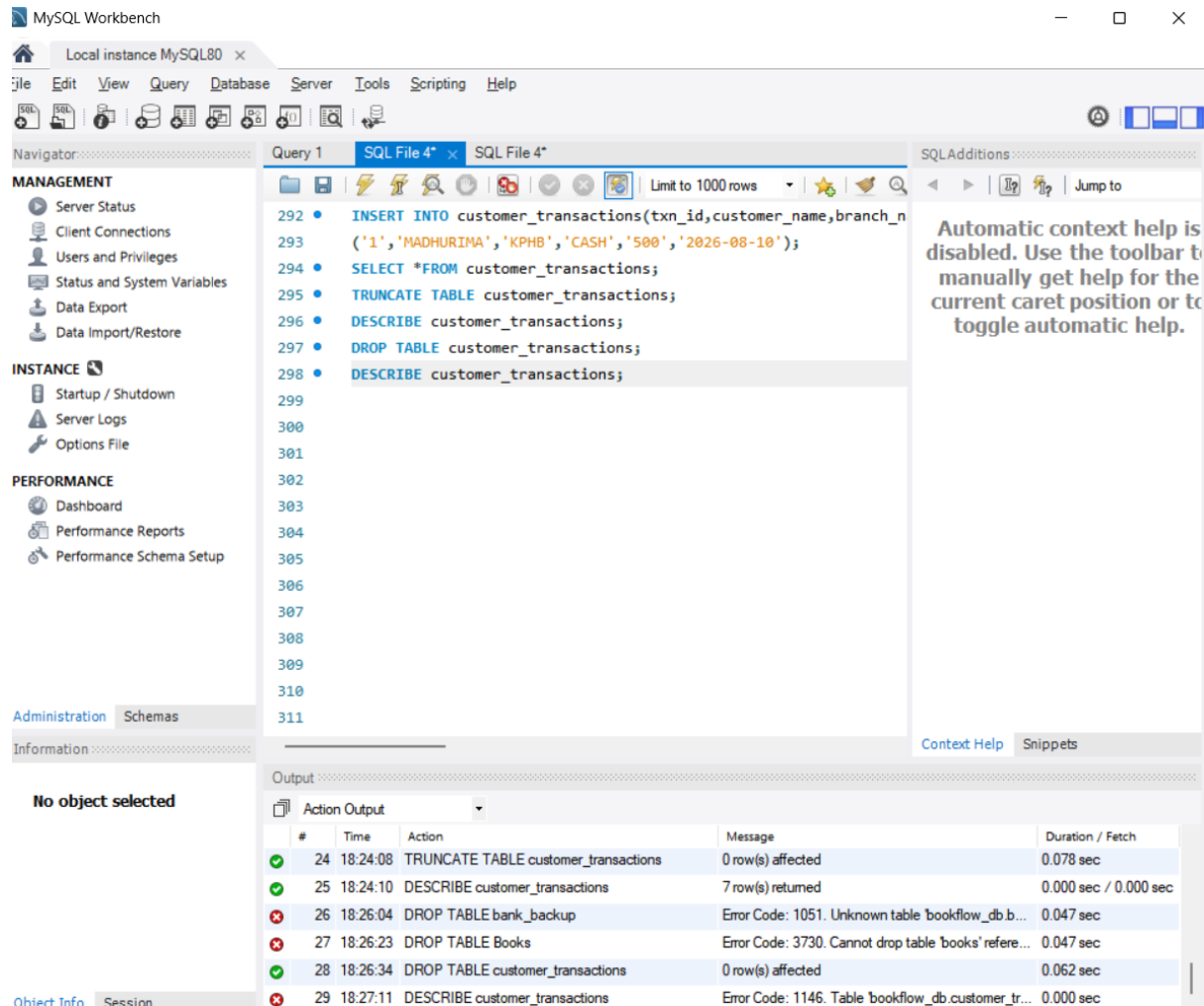
Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	18:22:11	DESCRIBE customer_transactions	7 row(s) returned	0.046 sec / 0.000 sec
21	18:22:40	SELECT *FROM customer_transactions LIMIT 0, ...	1 row(s) returned	0.000 sec / 0.000 sec
22	18:23:31	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.078 sec
23	18:23:33	DESCRIBE customer_transactions	7 row(s) returned	0.015 sec / 0.000 sec
24	18:24:08	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.078 sec
25	18:24:10	DESCRIBE customer_transactions	7 row(s) returned	0.000 sec / 0.000 sec

6.USING DROP TABLE

Customer_transactions table got deleted completed using DROP TABLE



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The main editor displays a SQL script with the following queries:

```
292 • INSERT INTO customer_transactions(txn_id,customer_name,branch_n
293 • ('1','MADHURIMA','KPHB','CASH','500','2026-08-10');
294 • SELECT *FROM customer_transactions;
295 • TRUNCATE TABLE customer_transactions;
296 • DESCRIBE customer_transactions;
297 • DROP TABLE customer_transactions;
298 • DESCRIBE customer_transactions;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 24	18:24:08	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.078 sec
✓ 25	18:24:10	DESCRIBE customer_transactions	7 row(s) returned	0.000 sec / 0.000 sec
✗ 26	18:26:04	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bookflow_db.b...	0.047 sec
✗ 27	18:26:23	DROP TABLE Books	Error Code: 3730. Cannot drop table 'books' refere...	0.047 sec
✓ 28	18:26:34	DROP TABLE customer_transactions	0 row(s) affected	0.062 sec
✗ 29	18:27:11	DESCRIBE customer_transactions	Error Code: 1146. Table 'bookflow_db.customer_tr...	0.000 sec

DML[DATA MANUPULATION LANGUAGE]

1.CREATED TABLE AND INSERTED VALUES

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator:.....

MANAGEMENT

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- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information:.....

No object selected

Object Info Session

Query 1 SQL File 4* x SQL File 4*

Limit to 1000 rows

```
314 (103,'ANU','Vijayawada','Deposit',12000,'2024-01-08'),
315 (104,'SARAT','Vizag','Deposit',8000,'2024-01-10'),
316 (105,'VIYAN','Hyderabad','Withdrawal',3500,'2024-01-11'),
317 (106,'RAJ','Vizag','Deposit',15000,'2024-01-12'),
318 (107,'CHARAN','Vijayawada','Withdrawal',1000,'2024-01-13'),
319 (108,'RAM','Hyderabad','Deposit',9000,'2024-01-14'),
320 (109,'SEETHARAM','Vizag','Withdrawal',4000,'2024-01-15'),
321 (110,'MADHURIMA','Vijayawada','Deposit',11000,'2024-01-16');
322 • SELECT *FROM bank_transactions;
```

Automatic disabled. Use manually g current care toggle au

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transac
▶	101	MADHU	Hyderabad	Deposit	5000.00	2024-01
	102	Sita	Hyderabad	Withdrawal	2000.00	2024-01
	103	ANU	Vijayawada	Deposit	12000.00	2024-01
	104	SARAT	ANU	Deposit	8000.00	2024-01
	105	VIYAN	Hyderabad	Withdrawal	3500.00	2024-01
	106	RAJ	Vizag	Deposit	15000.00	2024-01
	107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01
	108	RAM	Hyderabad	Deposit	9000.00	2024-01
	109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
	110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01

bank_transactions 10 x

Output

Action Output

#	Time	Action	Message
✓	28 18:26:34	DROP TABLE customer_transactions	0 row(s) affected
✗	29 18:27:11	DESCRIBE customer_transactions	Error Code: 1146. Table 'bookflow_db.customer_tr...'
✗	30 18:31:38	INSERT INTO bank_transactions VALUES (101,'...	Error Code: 1146. Table 'bookflow_db.bank_trans...
✓	31 18:32:20	CREATE TABLE bank_transactions (txn_id IN...	0 row(s) affected
✓	32 18:32:24	INSERT INTO bank_transactions VALUES (101,'...	10 row(s) affected Records: 10 Duplicates: 0 Wa...
✓	33 18:32:49	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned

2.INSERT

Added Krishna into table

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 4* x SQL File 4* SQL Addition:

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Object Info Session

```
318 (107,'CHARAN','Vijayawada','Withdrawal',1000,'2024-01-13'),
319 (108,'RAM','Hyderabad','Deposit',9000,'2024-01-14'),
320 (109,'SEETHARAM','Vizag','Withdrawal',4000,'2024-01-15'),
321 (110,'MADHURIMA','Vijayawada','Deposit',11000,'2024-01-16');
322 • SELECT *FROM bank_transactions;
323 • INSERT INTO bank_transactions
324 VALUES
325 (111,'KRISHNA','Vizag','Deposit',7000,'2024-01-18');
326 • SELECT *FROM bank_transactions;
```

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transac
	104	SARAT	Vizag	Deposit	8000.00	2024-01
	105	VIYAN	Hyderabad	Withdrawal	3500.00	2024-01
	106	RAJ	Vizag	Deposit	15000.00	2024-01
	107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01
	108	RAM	Hyderabad	Deposit	9000.00	2024-01
	109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
	110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01
	111	KRISHNA	Vizag	Deposit	7000.00	2024-01
*	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 11 x Apply Context Help

Output

Action Output

#	Time	Action	Message
30	18:31:38	INSERT INTO bank_transactions VALUES (101,'...'	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist
31	18:32:20	CREATE TABLE bank_transactions (txn_id IN...	0 row(s) affected
32	18:32:24	INSERT INTO bank_transactions VALUES (101,'...'	10 row(s) affected Records: 10 Duplicates: 0 V
33	18:32:49	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
34	18:34:58	INSERT INTO bank_transactions VALUES (111,'...'	1 row(s) affected
35	18:35:17	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned

3.UPDATE

Value is updated for 105 id

base Server Tools Scripting Help

Query 1 SQL File 4* x SQL File 4*

Limit to 1000 rows

```

324 VALUES
325 (111,'KRISHNA','Vizag','Deposit',7000,'2024-01-18');
326 • SELECT *FROM bank_transactions;
327
328 • UPDATE bank_transactions
329 SET amount = 5000
330 WHERE txn_id = 105;
331 • SELECT *FROM bank_transactions;
332

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transac
104	SARAT	Vizag	Deposit	8000.00	2024-01
105	VIYAN	Hyderabad	Withdrawal	5000.00	2024-01
106	RAJ	Vizag	Deposit	15000.00	2024-01
107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01
108	RAM	Hyderabad	Deposit	9000.00	2024-01
109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01
111	KRISHNA	Vizag	Deposit	7000.00	2024-01
*	NULL	NULL	NULL	NULL	NULL

bank_transactions 12 x Apply Revert Co

Output

Action Output

#	Time	Action	Message
✓ 32	18:32:24	INSERT INTO bank_transactions VALUES (101,'...	10 row(s) affected Records: 10 Dup
✓ 33	18:32:49	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
✓ 34	18:34:58	INSERT INTO bank_transactions VALUES (111,'...	1 row(s) affected
✓ 35	18:35:17	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned
✓ 36	18:36:46	UPDATE bank_transactions SET amount = 5000 ...	1 row(s) affected Rows matched: 1
✓ 37	18:36:57	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned

4.DELETE

DELETED 111 ID

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 4* x SQL File 4*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: No object selected

Query 1

```

327
328 • UPDATE bank_transactions
329 SET amount = 5000
330 WHERE txn_id = 105;
331 • SELECT *FROM bank_transactions;
332 • DELETE FROM bank_transactions
333 WHERE txn_id = 111;
334 • SELECT *FROM bank_transactions;
335

```

Limit to 1000 rows

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transac
103	ANU	Vijayawada	Deposit	12000.00	2024-01
104	SARAT	Vizag	Deposit	8000.00	2024-01
105	VIYAN	Hyderabad	Withdrawal	5000.00	2024-01
106	RAJ	Vizag	Deposit	15000.00	2024-01
107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01
108	RAM	Hyderabad	Deposit	9000.00	2024-01
109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01
NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 13 x Apply Revert Cont

Output

Action Output

#	Time	Action	Message
35	18:35:17	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned
36	18:36:46	UPDATE bank_transactions SET amount = 5000 ...	1 row(s) affected Rows matched: 1 Ch
37	18:36:57	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned
38	18:42:07	DELETE FROM bank_transactions WHERE txn_i...	1 row(s) affected
39	18:42:10	DELETE FROM bank_transactions WHERE txn_i...	0 row(s) affected
40	18:42:13	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned

Object Info Session

5.SELECT WITH WHERE CLAUSE

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 4* x SQL File 4*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: bank_transactions 14 x Apply Revert

No object selected

Object Info Session

```

331 • SELECT *FROM bank_transactions;
332 • DELETE FROM bank_transactions
333 WHERE txn_id = 111;
334 • SELECT *FROM bank_transactions;
335 • SELECT *
336 FROM bank_transactions
337 WHERE transaction_type='Deposit';
338
339

```

Result Grid Filter Rows: Edit: Export/Import

	txn_id	customer_name	branch_name	transaction_type	amount	transac
▶	101	MADHU	Hyderabad	Deposit	5000.00	2024-01
	103	ANU	Vijayawada	Deposit	12000.00	2024-01
	104	SARAT	Vizag	Deposit	8000.00	2024-01
	106	RAJ	Vizag	Deposit	15000.00	2024-01
	108	RAM	Hyderabad	Deposit	9000.00	2024-01
	110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01
*	NULL	NULL	NULL	NULL	NULL	NULL

Output

Action Output

#	Time	Action	Message
✓ 36	18:36:46	UPDATE bank_transactions SET amount = 5000 ...	1 row(s) affected Rows ma
✓ 37	18:36:57	SELECT *FROM bank_transactions LIMIT 0, 1000	11 row(s) returned
✓ 38	18:42:07	DELETE FROM bank_transactions WHERE txn_i...	1 row(s) affected
✓ 39	18:42:10	DELETE FROM bank_transactions WHERE txn_i...	0 row(s) affected
✓ 40	18:42:13	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
✓ 41	18:46:23	SELECT * FROM bank_transactions WHERE tra...	6 row(s) returned

6.DESC [SELECT WITH ORDER BY]

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 SQL File 4* x SQL File 4*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information: bank_transactions 15 x

Query 1

```

335 • SELECT *
336 FROM bank_transactions
337 WHERE transaction_type='Deposit';
338
339 • SELECT *
340 FROM bank_transactions
341 ORDER BY amount DESC;
342
343

```

Limit to 1000 rows

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transac
▶	106	RAJ	Vizag	Deposit	15000.00	2024-01
	103	ANU	Vijayawada	Deposit	12000.00	2024-01
	110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01
	108	RAM	Hyderabad	Deposit	9000.00	2024-01
	104	SARAT	Vizag	Deposit	8000.00	2024-01
	101	MADHU	Hyderabad	Deposit	5000.00	2024-01
	105	VIYAN	Hyderabad	Withdrawal	5000.00	2024-01
	109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
	102	Sita	Hyderabad	Withdrawal	2000.00	2024-01
	107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01

Result Grid Form Editor Field Types

Apply Revert

Output

Action Output

#	Time	Action	Message
✓	37 18:36:57	SELECT * FROM bank_transactions LIMIT 0, 1000	11 row(s) returned
✓	38 18:42:07	DELETE FROM bank_transactions WHERE txn_i...	1 row(s) affected
✓	39 18:42:10	DELETE FROM bank_transactions WHERE txn_i...	0 row(s) affected
✓	40 18:42:13	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
✓	41 18:46:23	SELECT * FROM bank_transactions WHERE tra...	6 row(s) returned
✓	42 18:47:25	SELECT * FROM bank_transactions ORDER BY ...	10 row(s) returned

No object selected

DCL[DATA CONTROL COMMANDS]

1. GRANT SELECT

MySQL Workbench

The screenshot displays the MySQL Workbench interface for a local instance of MySQL 8.0. The left sidebar contains navigation panels for MANAGEMENT, INSTANCE, and PERFORMANCE. The central editor shows a SQL script with the following queries:

```
349 • SELECT User, Host
350 FROM mysql.user
351 WHERE User = 'Auditor1';
352 • CREATE USER 'Auditor1'@'localhost'
353 IDENTIFIED BY 'password123';
354 • GRANT SELECT
355 ON bookflow_db.bank_transactions
356 TO 'Auditor1'@'localhost';
357 • SHOW GRANTS FOR 'Auditor1'@'localhost';
```

Below the editor, the 'Result Grid' shows the output of the queries. The first query returns the user and host information. The second query creates the user. The third query grants SELECT privileges on the bank_transactions table to the Auditor1 user. The fourth query shows the grants for the Auditor1 user.

The bottom panel shows the 'Output' window with a table of execution results:

#	Time	Action	Message
43	18:50:39	GRANT SELECT ON bank_transactions TO Audit...	Error Code: 1410. You are not allowed to c
44	18:51:53	GRANT SELECT ON bookflow_db.bank_transact...	Error Code: 1410. You are not allowed to c
45	18:52:19	SELECT User, Host FROM mysql.user WHERE U...	0 row(s) returned
46	18:52:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIE...	0 row(s) affected
47	18:52:42	GRANT SELECT ON bookflow_db.bank_transact...	0 row(s) affected
48	18:52:53	SHOW GRANTS FOR 'Auditor1'@'localhost'	2 row(s) returned

2. GRANT ALL

Granted all permissions to bank manager using grant all

MySQL Workbench

The screenshot displays the MySQL Workbench interface. The left sidebar contains navigation panels for 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'INSTANCE' (Startup / Shutdown, Server Logs, Options File), and 'PERFORMANCE' (Dashboard, Performance Reports, Performance Schema Setup). The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The main window shows a SQL script with the following queries:

```
359 • GRANT ALL
360 ON bank_transactions
361 TO BranchManager;
362 • CREATE USER 'BranchManager'@'localhost'
363 IDENTIFIED BY 'password123';
364 • GRANT ALL
365 ON bookflow_db.bank_transactions
366 TO 'BranchManager'@'localhost';
367 • SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Below the script, the 'Result Grid' shows the output of the 'SHOW GRANTS' query:

Grants for BranchManager@localhost	
▶	GRANT USAGE ON *,* TO 'BranchManager'@'...
	GRANT ALL PRIVILEGES ON 'bookflow_db', 'ba...

The bottom panel shows the 'Output' window with a table of execution results:

#	Time	Action	Message
✓ 47	18:52:42	GRANT SELECT ON bookflow_db.bank_transact...	0 row(s) affected
✓ 48	18:52:53	SHOW GRANTS FOR 'Auditor1'@'localhost'	2 row(s) returned
✗ 49	18:55:29	GRANT ALL ON bank_transactions TO BranchM...	Error Code: 1410. You are not allowed to cr
✓ 50	18:55:53	CREATE USER 'BranchManager'@'localhost' ID...	0 row(s) affected
✓ 51	18:56:00	GRANT ALL ON bookflow_db.bank_transactions ...	0 row(s) affected
✓ 52	18:56:09	SHOW GRANTS FOR 'BranchManager'@'localh...	2 row(s) returned

3.REVOKE SELECT

The screenshot shows the MySQL Workbench interface for a local instance of MySQL 8.0. The left sidebar contains navigation panels for MANAGEMENT, INSTANCE, and PERFORMANCE. The main editor displays a SQL script with the following queries:

```

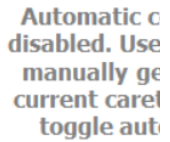
375
376
377 • SELECT User, Host
378 FROM mysql.user
379 WHERE User = 'Auditor1';
380 • SHOW GRANTS FOR 'Auditor1'@'localhost';
381 • REVOKE SELECT
382 ON bank_transactions
383 FROM 'Auditor1'@'localhost';
384
385
386
387
388
389
390
391
392
393
394

```

The bottom panel shows the Output window with the following table:

#	Time	Action	Message
52	18:56:09	SHOW GRANTS FOR 'BranchManager'@'localh...	2 row(s) returned
53	19:21:50	REVOKE SELECT ON bank_transactions FROM ...	Error Code: 1147. There is
54	19:22:37	SHOW GRANTS FOR 'Auditor1'@'%'	Error Code: 1141. There is
55	19:23:03	SELECT User, Host FROM mysql.user WHERE U...	1 row(s) returned
56	19:23:18	SHOW GRANTS FOR 'Auditor1'@'localhost'	2 row(s) returned
57	19:23:27	REVOKE SELECT ON bank_transactions FROM '...	0 row(s) affected

4.REVOKED ALL PERMISSIONS TO BANK MANAGER



TCL[TRANSACTION CONTROL LANGUAGE]

1.COMMIT

101 id deposite amount is updated from 5000 to 6000

The screenshot displays the MySQL Workbench interface. The left sidebar contains navigation panels for MANAGEMENT, INSTANCE, and PERFORMANCE. The central area shows a SQL query in the 'Query 1' tab, which is used to update the amount for transaction ID 101 and then commit the changes. The 'Result Grid' below the query shows the updated transaction details. The bottom panel shows the 'Output' tab with a log of the executed SQL statements and their results.

SQL Query:

```
392 • REVOKE ALL
393 • ON bank_transactions
394 • FROM 'BranchManager'@'localhost';
395 • UPDATE bank_transactions
396 • SET amount=6000
397 • WHERE txn_id=101;
398
399 • COMMIT;
400 • SELECT *FROM bank_transactions;
```

Result Grid:

txn_id	customer_name	branch_name	transaction_type	amount	transac
101	MADHU	Hyderabad	Deposit	6000.00	2024-01
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01
103	ANU	Vijayawada	Deposit	12000.00	2024-01
104	SARAT	Vizag	Deposit	8000.00	2024-01
105	VIYAN	Hyderabad	Withdrawal	5000.00	2024-01
106	RAJ	Vizag	Deposit	15000.00	2024-01
107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01
108	RAM	Hyderabad	Deposit	9000.00	2024-01
109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01
110	MADHURIMA	Vijayawada	Deposit	11000.00	2024-01

Output Log:

#	Time	Action	Message	Du
62	19:29:09	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.0
63	19:29:22	UPDATE bank_transactions SET amount=6000 ...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.0
64	19:29:27	UPDATE bank_transactions SET amount=6000 ...	0 row(s) affected Rows matched: 1 Changed: 0 ...	0.0
65	19:29:29	UPDATE bank_transactions SET amount=6000 ...	0 row(s) affected Rows matched: 1 Changed: 0 ...	0.0
66	19:29:32	COMMIT	0 row(s) affected	0.0
67	19:29:34	SELECT *FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.0

2.SAVE POINT

MySQL workbench

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' tab is selected, showing 'Schemas' and 'Information'. The main editor displays a SQL script with the following lines:

```
408
409
410 • START TRANSACTION;
411
412 • SAVEPOINT Before_Update;
413
414 • UPDATE bank_transactions
415   SET amount = 99999
416   WHERE txn_id = 102;
417
418 • ROLLBACK TO Before_Update;
419
420
421
422
423
424
425
426
427
```

The 'Output' tab at the bottom shows the execution results in a table:

#	Time	Action	Message
✓ 75	19:32:21	START TRANSACTION	0 row(s) affected
✓ 76	19:32:24	SAVEPOINT Before_Update	0 row(s) affected
✓ 77	19:32:26	UPDATE bank_transactions SET amount = 9999...	0 row(s) affected Rows matched: 1
✓ 78	19:32:27	UPDATE bank_transactions SET amount = 9999...	0 row(s) affected Rows matched: 1
✓ 79	19:32:29	UPDATE bank_transactions SET amount = 9999...	0 row(s) affected Rows matched: 1
✓ 80	19:32:30	ROLLBACK TO Before_Update	0 row(s) affected

3.FULL TCL EXAMPLE

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1 SQL File 4* x SQL File 4*

Limit to 1000 rows

```
416 WHERE txn_id = 102;
417
418 • ROLLBACK TO Before_Update;
419
420
421 • UPDATE bank_transactions
422 SET amount=7000
423 WHERE txn_id=101;
424
425 • SAVEPOINT SP1;
426
427 • UPDATE bank_transactions
428 SET amount=9000
429 WHERE txn_id=102;
430
431 • ROLLBACK TO SP1;
432
433 • COMMIT;
```

Output

Action Output

#	Time	Action	Message
✓ 84	19:33:53	SAVEPOINT SP1	0 row(s)
✓ 85	19:33:55	UPDATE bank_transactions SET amount=9000 ...	1 row(s)
✓ 86	19:33:58	UPDATE bank_transactions SET amount=9000 ...	0 row(s)
✓ 87	19:33:59	UPDATE bank_transactions SET amount=9000 ...	0 row(s)
✓ 88	19:34:01	ROLLBACK TO SP1	0 row(s)
✓ 89	19:34:04	COMMIT	0 row(s)